

Amazon EKS Control Plane & Managed Add-ons

Nexlify Technologies | Clusters: nexlify-prod-eks, nexlify-nonprod-eks | ap-south-1

1. Cluster Inventory & Version Policy

Nexlify operates two primary EKS clusters in ap-south-1. The production cluster 'nexlify-prod-eks' runs Kubernetes 1.30 (control plane) with node groups on 1.30. The non-production cluster 'nexlify-nonprod-eks' runs the same version for parity. Control plane upgrade cadence is every second Kubernetes minor release after the AWS-supported window opens; node AMI upgrades follow within 14 days. Upgrade windows are scheduled for Tuesday 02:00–04:00 IST. The control plane is multi-AZ by design (AWS managed). Endpoint access is private-only for production; non-prod allows public endpoint with CIDR restriction to the corporate VPN and GitHub Actions runner ranges. Cluster logging (API, audit, authenticator, controllerManager, scheduler) is enabled and shipped to CloudWatch Logs with 30-day retention.

2. Control Plane Configuration Baseline

Both clusters use the AWS-managed encryption configuration with a customer-managed KMS key for secrets encryption at rest. OIDC identity provider is enabled for IRSA (IAM Roles for Service Accounts). The cluster security group permits only necessary ports from the VPC CIDR and the shared-services VPC. Authentication is via AWS IAM Authenticator mapped to IAM Identity Center groups: eks-admins, eks-developers, eks-readonly. aws-auth ConfigMap is managed exclusively by Terraform; manual edits are overwritten. Control plane metrics are scraped by the in-cluster Prometheus and also available via CloudWatch Container Insights.

3. Required Managed Add-ons

The following EKS add-ons are installed and kept at the version recommended by the current Kubernetes release. Versions are pinned in Terraform and upgraded only after validation in non-prod:

- vpc-cni (amazon-k8s-cni) — custom networking disabled; prefix delegation enabled for higher pod density.
- coredns — two replicas minimum; autoscaling enabled via cluster-proportional-autoscaler in production.
- kube-proxy — iptables mode; IPVS considered for future high-churn workloads.
- aws-ebs-csi-driver — default StorageClass gp3 with encryption; volume expansion allowed.
- aws-efs-csi-driver — present only on non-prod for shared scratch volumes; not used in production.
- snapshot-controller — required for volume snapshot CRDs used by backup jobs.

4. Additional Platform Components

Beyond AWS managed add-ons, the platform deploys: Karpenter (see dedicated runbook), external-dns for Route 53 integration, cert-manager with Let's Encrypt and internal ACM private CA, metrics-server, cluster-autoscaler is intentionally not installed (Karpenter replaces it), and the AWS Load Balancer Controller for ALB/NLB provisioning. All platform components run in the 'platform-system' namespace with PodDisruptionBudgets and resource requests/limits defined. Helm charts are rendered by Argo CD from the platform-gitops repository; direct helm install is prohibited.

5. Control Plane Upgrade Procedure (Summary)

1. Announce upgrade window in #platform-ops and freeze non-critical changes. 2. Upgrade non-prod control plane first; monitor for 48 hours. 3. Upgrade add-ons to compatible versions. 4. Drain and replace nodes via Karpenter consolidation or explicit NodeClaim deletion. 5. Repeat for production after CAB approval. 6. Validate IRSA, ingress, and critical application health checks. Full procedure with rollback is in the detailed EKS upgrade playbook (NX-OPS-EKS-UPG-001).

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